Causality and lack thereof of multipolar polarizabilities

These notes concern the paper: Causality relations in the homogenization of metamaterials (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.054305?casa_token=D5NLdQGheXgAAAAA%3AIZHhRs0OZaJXAAsHsry1trpzxjyL95JlORfr_BnshrBMEFWlKWvJzg-3B9V9GgqYq_TfgZo091Wlgg)

Derivation of Equation 7

We start by proving **Equation 7**. This equation follows from taking $l = 1$ in **Equation 8** in the paper: Localized plasmons in graphene-coated nanospheres (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414?casa_token=0sofuUKZtWoAAAAA%3AtWtgtm58Ofc5KkLAiOfrI91Q6lSiNsQlxlZ-xs2OKa5YOKbLy2batjrduNASmGf1lidKyaO6ykOugw):

$$\alpha_1 = -\frac{4\pi(3!!)^2}{6i}\frac{1}{(\omega/c)^3}\frac{\varepsilon j_1(x)\Big[yj_1(y)\Big]' - j_1(y)\Big[xj_l(x)\Big]'}{\varepsilon j_1(x)\Big[yh_1^{(1)}(y)\Big]' - h_1^{(1)}(y)\Big[xj_l(x)\Big]'} = \\ -\frac{6\pi}{i(\omega/c)^3}\frac{u}{u+iv} = \frac{6\pi}{(\omega/c)^3}\frac{1}{(v/u)-i},$$

which is exactly **Equation 7** of the paper by **Andrea Alu** (note that the quantities u, v are defined in the **Alu** paper) and we have defined $x = \sqrt{\varepsilon}(a\omega/c) = \sqrt{\varepsilon}y$ in the above (where ε is the relative permittivity of the spherical scatterer). Note the use of the double factorial, $!!$ above. The double factorial is the same as the regular factorial except that one skips every other number (e.g. $3!! = 3 \times 1$)

Derivation of Equation 15

We derive **Equation 15** of the **Alu** paper from **Equation 9** of thesame paper (we derive **Equation 9** of the paper in the next section). **Equation 9** is given by:

$$\alpha_d^{-1} = a^{-3}\left[\frac{1}{4\pi}\frac{\varepsilon + 2\varepsilon_0}{\varepsilon - \varepsilon_0} - \frac{3(k_0 a)^2}{20\pi}\frac{\varepsilon - 2\varepsilon_0}{\varepsilon - \varepsilon_0} - i\frac{(k_0 a)^3}{6\pi}\right]$$

Note that we use the notation of the paper, which means that ε in this expression is not a relative permittivity but the full permittivity (i.e. $\varepsilon \to \varepsilon_0\varepsilon$). To arrive at **Equation 15**, we take the $(k_0 a) \to 0$ limit while maintaining the imaginary component. This means that we neglect the effect of the second term in the bracket:

$$\alpha_d^{-1} \approx \alpha_s^{-1} \equiv a^{-3}\left[\frac{1}{4\pi}\frac{\varepsilon+2\varepsilon_0}{\varepsilon-\varepsilon_0} - i\frac{(k_0 a)^3}{6\pi}\right] \to \alpha_s = a^3\left[\frac{1}{4\pi}\frac{\varepsilon+2\varepsilon_0}{\varepsilon-\varepsilon_0} - i\frac{(k_0 a)^3}{6\pi}\right]^{-1}$$
$$= a^3\left[\frac{(k_0 a)^3}{6\pi}\left[\frac{3}{2(k_0 a)^3}\frac{\varepsilon+2\varepsilon_0}{\varepsilon-\varepsilon_0} - i\right]\right]^{-1} = \frac{6\pi}{k_0^3}\left[(k_0 a x)^{-3} - i\right]^{-1},$$

where x is defined in the paper. The above is exactly **Equation 15** of the paper.

Derivation of Equation 9

Equation 9 of the **Alu** paper is the quasistatic approximation of the polarizability of a sphere, taking into account radiation losses. In order to derive this, we use the expansion of the spherical Bessel functions for small arguments (**Equations 5.1, 5.2** in **Absorption and Scattering of Light by Small Particles** by **Bohren and Huffman**). We do this to second order in the dimensionless quantity, $k_0 a \equiv a\omega/c$, for each term containing a Bessel function (as per what is stated in the **Alu** paper). Our approximations are, explicitly,

$$j_1(\rho) \approx \frac{\rho}{3}\left[1 - \frac{\rho^2}{10}\right] \to (\rho j_1 \rho)' \approx \frac{2\rho}{3}\left[1 - \frac{\rho^2}{5}\right]$$

$$y_1(\rho) \approx -\frac{1}{\rho^2}\left[1 + \frac{\rho^2}{2}\right] \to (\rho y_1(\rho))' \approx \frac{1}{\rho^2}\left[1 - \frac{\rho^2}{2}\right]$$

From which we also have:

$$j_1(\rho_1)\left[\rho_2 j_1(\rho_2)\right]' \approx \rho_1\rho_2\left[\frac{2}{9} - \frac{1}{45}\rho_1^2 - \frac{2}{45}\rho_2^2\right]$$

$$j_1(\rho_1)\left[\rho_2 y_1(\rho_2)\right]' \approx \frac{\rho_1}{\rho_2^2}\left[\frac{1}{3} - \frac{\rho_2^2}{6} - \frac{\rho_1^2}{30}\right]$$

$$y_1(\rho_2)\left[\rho_1 j_1(\rho_1)\right]' \approx -\frac{\rho_1}{\rho_2^2}\left[\frac{2}{3} - \frac{2\rho_1^2}{15} + \frac{\rho_2^2}{3}\right]$$

With these approximations, we may find approximate expressions for u and v (quantities defined in the **Alu** paper):

$$\varepsilon\varepsilon_0 u \equiv \varepsilon j_1(ka)[(k_0a)j_1(k_0a)]' - \varepsilon_0 j_1(k_0a)[(ka)j_1(ka)]'$$

$$(ka)(k_0a)\left[\frac{2(\varepsilon-\varepsilon_0)}{9} - \frac{(\varepsilon-2\varepsilon_0)(ka)^2}{45} - \frac{(2\varepsilon-\varepsilon_0)(k_0a)^2}{45}\right] = (ka)(k_0a)\left[\frac{2(\varepsilon-\varepsilon_0)}{9} - \frac{\varepsilon^2-\varepsilon_0^2}{45\varepsilon_0}(k_0a)^2\right]$$

$$(ka)(k_0a)\frac{2(\varepsilon-\varepsilon_0)}{9}\left[1 - \frac{(\varepsilon+\varepsilon_0)}{10\varepsilon_0}(k_0a)^2\right.$$

where, as stated in the **Alu** paper, $k \equiv k_0\sqrt{\varepsilon/\varepsilon_0}$ is the momentum of the light in the spherical scatterer.

Similarly, we have, for v:

$$\varepsilon\varepsilon_0 v \equiv \varepsilon j_1(ka)[(k_0a)y_1(k_0a)]' - \varepsilon_0 y_1(k_0a)[(ka)j_1(ka)]' \approx$$
$$\left[\frac{\varepsilon+2\varepsilon_0}{3} - \frac{(\varepsilon-2\varepsilon_0)(k_0a)^2}{6} - \frac{(\varepsilon+4\varepsilon_0)(ka)^2}{30}\right]\frac{(ka)}{(k_0a)^2} =$$
$$\frac{(ka)}{3(k_0a)^2}\left[(\varepsilon+2\varepsilon_0) - \frac{5(\varepsilon-2\varepsilon_0)\varepsilon_0 + (\varepsilon+4\varepsilon_0)\varepsilon}{10\varepsilon_0}(k_0a)^2\right] =$$
$$\frac{(ka)}{(k_0a)^2}\frac{\varepsilon+2\varepsilon_0}{3}\left[1 - \frac{5(\varepsilon-2\varepsilon_0)\varepsilon_0 + \varepsilon^2 + 4\varepsilon\varepsilon_0}{10\varepsilon_0(\varepsilon+2\varepsilon_0)}(k_0a)^2\right] =$$
$$\frac{(ka)}{(k_0a)^2}\frac{\varepsilon+2\varepsilon_0}{3}\left[1 - \frac{\varepsilon^2 + 9\varepsilon\varepsilon_0 - 10\varepsilon_0^2}{10\varepsilon_0(\varepsilon+2\varepsilon_0)}(k_0a)^2\right]$$

We have that u, v are of the forms:

$$v = \frac{(ka)}{(k_0a)^2}\frac{\varepsilon+2\varepsilon_0}{3}\left[1 - p(k_0a)^2\right], u = (ka)(k_0a)\frac{2(\varepsilon-\varepsilon_0)}{9}\left[1 - r(k_0a)^2\right]$$

Which means that the ratio, v/u will be given by (to next to lowest order):

$$(v/u) \approx \frac{3(\varepsilon+2\varepsilon_0)}{2(\varepsilon-\varepsilon_0)}\frac{1}{(k_0a)^3}\left[1+(r-p)(k_0a)^2\right]$$

Where

$$r-p=\frac{(\varepsilon+2\varepsilon_0)(\varepsilon+\varepsilon_0)-\varepsilon^2-9\varepsilon\varepsilon_0+10\varepsilon_0^2}{10\varepsilon_0(\varepsilon+2\varepsilon_0)}=\frac{-6\varepsilon+12\varepsilon_0}{10(\varepsilon+2\varepsilon_0)}=-\frac{3}{5}\left[\frac{\varepsilon-2\varepsilon_0}{\varepsilon+2\varepsilon_0}\right]$$

Therefore, we have that the ratio, v/u, is approximately given by:

$$(v/u) \approx \frac{3(\varepsilon+2\varepsilon_0)}{2(\varepsilon-\varepsilon_0)}\frac{1}{(k_0a)^3}\left[1-\frac{3}{5}\frac{\varepsilon-2\varepsilon_0}{\varepsilon+2\varepsilon_0}(k_0a)^2\right]$$

From which we obtain:

$$\alpha_d^{-1}=\frac{(\omega/c)^3}{6\pi}(v/u-i)=\frac{k_0^3}{6\pi}\left[\frac{3}{2}\frac{\varepsilon+2\varepsilon_0}{\varepsilon-\varepsilon_0}\frac{1}{(k_0a)^3}-\frac{9}{10}\frac{(\varepsilon-2\varepsilon_0)}{(\varepsilon-\varepsilon_0)}\frac{(k_0a)^2}{(k_0a)^3}-i\right]=$$
$$\frac{1}{a^3}\left[\frac{1}{4\pi}\frac{\varepsilon+2\varepsilon_0}{\varepsilon-\varepsilon_0}-\frac{3}{20\pi}\frac{\varepsilon-2\varepsilon_0}{\varepsilon-\varepsilon_0}(k_0a)^2-i\frac{(k_0a)^3}{6\pi}\right],$$

which is exactly **Equation 9** in the **Alu** paper.

Quasistatic polarizability in the time domain (Equation 16)

We wish to evaluate the quasistatic polarizability in the time domain to show that it is not strictly causal (i.e. it contains a non-zero component proportional to $\Theta(-t)$ in addition to usual causal component, which is proportional to $\Theta(t)$ (where Θ is the usual notation for the Heaviside step function)). The quasistatic polarizability in the time domain is given by the Fourier transform:

$$\alpha_{st}(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\alpha_s(\omega)e^{-i\omega t}\mathrm{d}\omega,$$

where (following **Equation 15** of the **Alu** paper and using their notation),

$$\alpha_s(\omega)=\frac{6\pi}{(\omega/c)^3}\frac{(xa\omega/c)^3}{1-i(xa\omega/c)^3}=\frac{6\pi i x^3a^3}{(xa\omega/c)^3+i}$$

In order to find the Fourier transform of this function, we note that it may be written as a product of simple poles. These poles are the cubic roots of $e^{-i\pi/2}$, given by $e^{-i\pi/6} = \sqrt{3}/2 - i/2, e^{-i\pi/6+2\pi i/3} = e^{i\pi/2} = i, e^{-i\pi/6-2\pi i/3} = e^{-5\pi i/6} = -\sqrt{3}/2 - i/2$, meaning we may write the denominator in the form:

$$\frac{1}{(xa\omega/c)^3 + i} = \frac{1}{((xa\omega/c) - e^{-i\pi/6})((xa\omega/c) - e^{-5i\pi/6})((xa\omega/c) - i)}$$

Therefore, we have that the Fourier transform is given by (after taking a u-substitution $\omega \to \omega c/xa$, which preserves the integration limits):

$$\alpha_{st}(t) = \frac{1}{2\pi}(6\pi i x^3 a^3)\frac{c}{xa}\int_{-\infty}^{\infty} e^{-i\omega ct/xa}\left[\frac{1}{(\omega - e^{-i\pi/6})(\omega - e^{-5i\pi/6})(\omega - i)}\right]\mathrm{d}\omega$$

For $t < 0$, we close the contour in the upper half of the complex plane, since $e^{-i\omega ct/xa}$ vanishes at infinity if the imaginary part of ω is positive for $t < 0$. Therefore, integrating over a semicircle contour will just have a contribution at the pole at $\omega = i$. This gives us:

$$\alpha_{st}(t < 0) = -6\pi x^2 a^2 c e^{ct/xa}\left[\frac{1}{(i - \sqrt{3}/2 + i/2)(i + \sqrt{3}/2 + i/2)}\right] = 2\pi x^2 c a^2 e^{ct/xa},$$

which is exactly the term proportional to $\Theta(-t)$ in **Equation 16** of the **Alu** paper.
For $t > 0$, we close the contour in the lower half of the complex plane, since that is where $e^{-i\omega ct/xa}$ goes to zero at negative imaginary infinity. This time, we get contributions from two simple poles:

$$\alpha_{st}(t>0) = 6\pi x^2 a^2 c e^{-ct/2xa}\left[e^{-\sqrt{3}cti/2xa}\left[\frac{1}{(e^{-i\pi/6}-e^{-5\pi i/6})(e^{-i\pi/6}-i)}\right]+ e^{\sqrt{3}cti/2xa}\left[\frac{1}{(e^{-5\pi i/6}-e^{-i\pi/6})(e^{-5\pi i/6}-i)}\right]\right] = \frac{6\pi x^2a^2ce^{-ct/2xa}}{3}\left[\frac{e^{-\sqrt{3}cti/2xa}}{1/2-\sqrt{3}i/2}+\frac{e^{\sqrt{3}cti/2xa}}{1/2+\sqrt{3}i/2}\right] = 2\pi x^2a^2ce^{-ct/2xa}\left[\cos(\sqrt{3}ct/2xa)+\sqrt{3}\sin(\sqrt{3}ct/2xa)\right],$$

which is equal to the term proportional to $\Theta(t)$ in **Equation 16** of the **Alu** paper.

Derivation of Equation 20

Equation 20 is the generalized **Kramers-Kronig** relations for the polarizability. The polarizability is not strictly causal, so one has to add a correction to the regular **Kramers-Kronig** relations to account for the contribution from the non-causal component.

We proceed as follows: by definition, the Fourier transform of $\alpha(t)$ is given by:

$$\alpha(\omega) = \int_{-\infty}^{\infty} e^{i\omega t}\alpha(t)\mathrm{d}t = \int_{-\infty}^{\infty}\Theta(t)\alpha(t)e^{i\omega t}\mathrm{dt} + \int_{-\infty}^{0}\alpha(t)e^{i\omega t}\mathrm{d}t,$$

where, as before, $\Theta(t)$ is the Heaviside step function. We now expand the Heaviside function in Fourier components to write:

$$\Theta(t) = \frac{i}{2\pi}\int_{-\infty}^{\infty}\frac{e^{-i\omega t}}{\omega+i\delta}\mathrm{d}\omega \to \int_{-\infty}^{\infty}\Theta(t)\alpha(t)e^{i\omega t}\mathrm{d}t = \frac{i}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{e^{i(\omega-\Omega)t}}{\Omega+i\delta}\alpha(t)\mathrm{dtd}\Omega = \frac{i}{2\pi}\int_{-\infty}^{\infty}\frac{\alpha(\omega-\Omega)}{\Omega+i\delta}\mathrm{d}\Omega = \frac{i}{2\pi}\int_{-\infty}^{\infty}\frac{\alpha(\Omega)}{\omega-\Omega+i\delta}\mathrm{d}\Omega$$

In the last step, above, we made a change of variables $\Omega' = \omega - \Omega$. From the above, we may therefore write (using the fact that $1/(a+i\delta) = -i\pi\delta(a) + P(a)^{-1}$, with P denoting Cauchy principal value):

$$\alpha(\omega) = \frac{1}{2}\alpha(\omega) + \frac{i}{2\pi}P\int_{-\infty}^{\infty}\frac{\alpha(\Omega)}{\omega-\Omega}\mathrm{d}\Omega + \int_{-\infty}^{0}\alpha(t)e^{i\omega t}\mathrm{dt}$$

We take the real part of this equation (and use the fact that $\alpha^*(\omega) = \alpha(-\omega)$ to restrict the frequency integral to positive frequencies) to give us:

$$\alpha'(\omega) = -\frac{1}{\pi} P \int_0^\infty \alpha''(\Omega) \left[\frac{1}{\omega - \Omega} - \frac{1}{\omega + \Omega}\right] \mathrm{d}\Omega + 2\int_{-\infty}^0 \alpha(t)\cos(\omega t)\mathrm{d}t = \\ -\frac{2}{\pi} P \int_0^\infty \alpha''(\Omega) \frac{\Omega}{\omega^2 - \Omega^2} \mathrm{d}\Omega + 2\int_{-\infty}^0 \alpha(t)\cos(\omega t)\mathrm{d}t,$$

which is **Equation 20** of the paper.

Derivation of Equation 21

Equation 21 is the quasistatic approximation to **Equation 20**. Note that although the paper restricts the **Kramers-Kronig** correction to the region $-2a/c < t < 0$, this is not strictly correct for the quasistatic corection. This is because although the fully retarded polarizability is only non-zero for $t > -2a/c$, the quasistatic polarizability has an exponential tail that persists below $t = -2a/c$. With this in mind, the quasistatic correction is given by:

$$4\pi x^2 ca^2 \int_{-\infty}^0 e^{ct/xa} \cos(\omega t)\mathrm{d}t = 2\pi x^2 ca^2 \int_{-\infty}^0 \left[e^{t(c/xa + i\omega)} + e^{t(c/xa - i\omega)}\right] \mathrm{d}t = \\ 2\pi x^2 a^2 c \left[\frac{1}{c/xa + i\omega} + \frac{1}{c/xa - i\omega}\right] = \frac{4\pi x a c^2}{c^2/x^2a^2 + \omega^2} = \frac{4\pi x^3 a^3 c^2}{c^2 + \omega^2 x^2 a^2},$$

which is precisely the correction shown in **Equation 21**.